

MITO, Japan

WMO: 476290

Lat: 36.3811N

Lon: 140.4681E

Elev: 102

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	27.0	9.3	8.9	40.1	12.2	10.3	39.2	18.6	42.3	15.8	42.5	3.4	340	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	91.7	78.3	89.1	77.5	86.5	76.6	79.6	88.3	78.4	86.2	77.4	84.3	7.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	142.2	83.8	76.3	137.7	82.8	75.4	133.5	81.9	43.2	88.7	42.1	86.6	41.0	84.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.0	13.5	11.7	DB	21.0	95.9	1.9	2.1	19.7	97.4	18.5	98.6	17.4	99.7	16.0	101.3
			WB	19.5	81.7	1.9	1.2	18.1	82.5	17.0	83.2	16.0	83.9	14.6	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	38.4	39.5	45.2	54.3	63.0	68.9	76.6	79.0	72.5	62.4	51.7	42.6
	DBStd	14.89	3.87	4.61	5.65	6.02	5.14	4.92	5.41	4.63	5.59	5.33	5.43	4.95
	HDD50	1134	360	295	173	26	0	0	0	0	0	0	44	236
	HDD65	3802	825	713	613	325	98	17	0	0	6	114	398	693
	CDD50	4043	0	2	25	154	403	568	826	900	675	386	97	7
	CDD65	1235	0	0	0	3	36	135	361	435	231	34	0	0
	CDH74	9004	0	0	0	26	198	608	2813	3873	1371	115	0	0
	CDH80	2871	0	0	0	1	24	123	979	1384	343	17	0	0
Wind	WSAvg	5.0	4.6	5.1	5.7	6.0	5.5	5.2	5.1	5.1	5.1	4.7	4.1	4.3
Precipitation	PrecAvg	54.1	2.0	2.0	3.4	4.6	5.6	5.7	5.9	5.1	7.6	6.9	3.3	2.0
	PrecMax	74.8	4.8	6.7	5.9	8.8	13.3	9.8	12.6	15.5	16.2	20.0	8.1	7.4
	PrecMin	43.7	0.2	0.3	1.6	1.1	3.0	2.5	1.2	0.8	0.4	0.5	0.0	0.0
	PrecStd	7.0	1.3	1.4	1.3	2.1	2.1	1.7	2.8	3.4	3.6	4.4	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	63.3	69.7	77.7	83.4	87.3	93.9	95.0	90.1	82.3	70.3	63.2
		MCWB	48.0	53.9	54.9	60.3	66.6	74.0	78.2	79.8	76.2	71.4	60.2	55.5
	2%	DB	53.1	55.9	64.6	72.6	79.0	83.6	91.5	92.3	87.0	76.8	67.1	58.7
		MCWB	43.7	46.4	53.6	60.0	65.2	71.9	78.0	78.9	76.1	67.5	58.4	50.1
	5%	DB	50.1	52.3	60.6	69.0	75.7	80.2	88.7	89.8	84.0	73.9	64.5	55.6
		MCWB	40.8	43.2	51.4	58.2	63.7	70.1	77.5	78.2	75.2	65.9	56.7	47.7
	10%	DB	47.6	49.1	56.9	65.7	72.9	77.3	85.6	87.3	81.0	71.3	62.0	52.7
		MCWB	39.3	41.0	49.1	56.2	62.9	69.4	76.4	77.4	73.8	63.8	55.4	45.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	56.0	59.1	67.4	71.5	76.4	80.7	81.7	78.9	73.3	63.9	58.4
		MCDB	55.5	61.7	63.2	72.4	77.7	84.7	90.3	91.7	85.3	78.2	67.4	60.9
	2%	WB	45.6	48.4	56.3	62.8	68.3	74.3	79.2	80.1	77.4	70.2	61.1	52.6
		MCDB	50.4	53.4	62.3	68.8	74.7	80.7	88.4	89.0	83.5	74.8	64.4	56.6
	5%	WB	42.6	45.1	53.3	59.9	66.4	72.4	78.0	79.0	76.4	67.9	59.0	49.1
		MCDB	48.3	50.3	59.2	66.1	72.7	77.9	86.1	87.1	82.2	72.0	62.5	53.8
	10%	WB	40.2	42.2	49.6	57.8	64.6	70.4	76.9	78.1	75.1	65.5	56.7	46.5
		MCDB	46.1	47.6	55.2	63.7	70.1	75.4	83.9	85.2	80.5	69.9	60.8	51.3
Mean Daily Temperature Range	5% DB	MDBR	18.5	18.0	18.1	17.9	15.5	12.2	12.4	12.7	12.7	14.5	17.3	18.3
		MCDBR	21.9	22.5	24.4	23.8	21.2	17.4	17.1	16.9	15.5	16.5	19.6	20.9
		MCWBR	13.8	14.8	15.5	13.3	10.7	8.1	6.8	6.4	6.9	9.3	12.6	13.8
	5% WB	MCDBR	19.6	20.5	21.9	19.3	16.9	14.4	15.7	15.2	14.2	13.5	16.0	18.1
		MCWBR	14.1	15.4	16.4	12.7	10.2	7.7	6.8	6.4	6.9	9.3	12.2	13.8
Clear-Sky Solar Irradiance	taub	0.372	0.422	0.495	0.548	0.568	0.603	0.598	0.582	0.511	0.471	0.418	0.376	
	taud	2.247	2.098	1.921	1.846	1.852	1.836	1.899	1.930	2.087	2.116	2.182	2.256	
	Ebn at Noon	253	252	243	238	236	227	227	227	237	233	233	242	
	Edh at Noon	33	43	57	64	65	66	62	59	48	43	36	31	
All-Sky Solar Radiation	RadAvg	862	1034	1288	1508	1621	1482	1540	1516	1198	955	785	746	
	RadStd	47	95	71	134	159	115	242	163	96	90	69	49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page